

Numerical Solution of the Time-Dependent Schrödinger Equation with Coulomb Interactions

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Abstract

We survey a number of problems in atomic physics which reduce to the solution of the time-dependent Schrödinger equation in two or three dimensions. Direct numerical techniques are discussed in two stages, the construction of propagators and discretization in space. Under the latter heading, finite element methods are emphasized.

1. Introduction: Applications of the Time-Dependent Schrödinger Equation

A. General Remarks

The time-dependent Schrödinger equation of a system whose Hamilton is H ,

$$\left[H(q,p,t) - i \frac{\partial}{\partial t} \right] \Phi(q,t) = 0 \quad (1)$$

uniquely specifies the evolution of the wavefunction Φ for all $t > 0$ if its value at $t = 0$ is given.* For a system of particles interacting through local potentials, Eq. (1) may be characterized mathematically as a diffusion equation in imaginary time with a diffusion coefficient variable in space (to account for the potential energy). One should be aware that not all statements about diffusion in real time carry over, e.g., stability criteria are less severe.

For half a century Eq. (1) has been solved by the eigenfunction expansion method initiated by Dirac [1] which for many problems is entirely adequate. However, in other problems, notably where more than two particles are unbound, eigenfunction expansions encounter severe difficulties, both formally and practically [2] so that alternative approaches must be sought. Direct numerical solution, as in fluid mechanics, has made little headway until recently. Very beautiful results have been achieved in the fields of chemical [3] and nuclear reactions [4] using finite-difference methods. These are for the most part two-dimensional (2D) calculations; a solution in three dimensions (3D) is truly a tour de force! Even in 2D, large numbers of mesh points and very careful numerical analysis are required.

* Atomic units will be used throughout, so that $\hbar = 1$.

When Coulomb interactions are present, two additional difficulties appear. In most numerical methods the solutions are unstable where two charged particles are close together. Also, because the Coulomb interaction dies off so slowly, the wavefunction must be represented at very large separations of the particles. For these reasons, little progress has been made on the simplest atomic systems, $e + H$ and $p + H$. One note on head-on $p + H$ collisions has been published [5]. Three unbound charged particles exhibit a number of phenomena not found in other fields of physics, notably exotic threshold laws [6] in the $e + e + p$ system, and capture-into-the-continuum in the $e + p + p$ systems [7]. It is obvious that the number of three-body systems is greatly multiplied if bare nuclei other than protons are considered [8], not to mention such exotics as positrons, muons, mesons, etc.

In this article we shall propose the introduction of finite element methods as a way of solving problems with Coulomb interactions. This section will be concluded by setting up the Hamiltonian for some representative problems. Following sections deal with numerical algorithms.

B. Proton-Hydrogen Collisions

The Hamiltonian for $p + H$, in the energy regime where the nuclei can be treated classically ($E > 500$ eV), is given by

$$H = -\frac{1}{2} \nabla^2 - \frac{1}{r} - \frac{1}{|\mathbf{r} - \mathbf{R}(t)|} \quad (2)$$

where $\mathbf{r}$ is the position of the electron relative to one of the nuclei as origin and $\mathbf{R}$ is the position of the other nucleus. The nuclei can be taken to move uniformly in straight lines. To reduce Eq. (2) to 2D, one can transform into a set of coordinates rotating with $\mathbf{R}$ and discard the Coriolis terms. Then if ρ, z, ϕ are cylindrical coordinates the Hamiltonian is independent of ϕ ,

$$H' = -\frac{1}{2} \left(\frac{\partial^2}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial}{\partial \rho} + \frac{\partial^2}{\partial z^2} \right) - \frac{1}{r_A} - \frac{1}{r_B} \quad (3)$$

$$r_A^2 = \rho^2 + z^2, r_B^2 = \rho^2 + (z - R)^2, R^2 = b^2 + v^2 t^2$$

The transformation to Eq. (3) is of practical significance, since the Coriolis terms appear to have little effect on total charge exchange and ionization cross sections [9]. However, caution must be observed since Eq. (2) leads to Galilean invariance, but Eq. (3) does not. In particular, if the incoming proton is switched off the bound electron described by H' will have transitions unless its nucleus is taken as the fixed origin. This last mentioned prescription does lead to sensible numerical results but a less empirical procedure should be sought.

Another reduction to 2D is obtained if the y -coordinate perpendicular to the collision plane is "frozen out" by making the ansatz

$$\psi_3(x,y,z,t) \cong \chi(y) \psi_2(x,z,t) \quad (4)$$

Regardless of its physical validity, this might provide an instructive model to investigate the neglect of coriolis coupling in Eq. (3).

C. Electron-Hydrogen Collisions

The Hamiltonian describing $e + e + H$ is stationary,

$$H = -\frac{1}{2}(\nabla_1^2 + \nabla_2^2) + \left(\frac{1}{r_{12}} - \frac{1}{r_1} - \frac{1}{r_2}\right) \quad (5)$$

where $\mathbf{r}_1$ and $\mathbf{r}_2$ are the coordinates of the two electrons, and the nucleus is regarded as infinitely massive. It is explained in textbooks [10] that the wavefunctions of a state of given angular momentum L and spin S may be expanded in symmetric top eigenfunctions of a set of Euler angles $\alpha\beta\gamma$

$$\psi_{SL} = \sum_K \Phi_K(r_1 r_2 \theta) D_{LKM}(\alpha\beta\gamma) \quad (6)$$

where θ is the angle between $\mathbf{r}_1$ and $\mathbf{r}_2$ and the symmetry of Φ is determined by S . In general, the Φ_K satisfy a set of coupled partial differential equations, but for $L = 0$ we have only to consider the effective Hamiltonian,

$$H' = \frac{1}{2} \left(\frac{\partial^2}{\partial r_1^2} + \frac{\partial^2}{\partial r_2^2} \right) - \frac{1}{2} \left(\frac{1}{r_1^2} + \frac{1}{r_2^2} \right) \frac{1}{\sin \theta} \frac{1}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \left(\frac{1}{r_{12}} - \frac{1}{r_1} - \frac{1}{r_2} \right) \quad (7)$$

We obtain a 2D model (associated with the same of Peterkop [11]) from Eq. (7) by setting $\theta = \pi$ so that the electrons are always diametrically opposed. Though apparently simple-minded such a model offers a great deal of physical insight and we chose it as a first test of our methods.

D. Many-Electron Systems: TDHF

As the number of electrons increases the dimensionality of Eq. (1) rapidly grows out of hand. However, an independent particle model would remain tractable, and the best one can do in this direction is to introduce a self-consistent field. Thus, we consider N electrons moving in an external potential

$$V = \sum_{m=1}^N v(m) \quad (8)$$

which might arise from nuclei in a heavy-particle collision problem. The single-particle orbitals satisfy (neglecting exchange)

$$\left(-\frac{1}{2} \nabla^2 + v + U \right) \phi_m = i \frac{\partial \phi_m}{\partial t} \quad (9)$$

where the self-consistent potential satisfies Poissons' equation,

$$\nabla^2 U = -4\pi \sum_{m=1}^N |\phi_m|^2 \quad (10)$$

This sets up time-dependent Hartree–Fock theory (TDHF) which is familiar to quantum chemists in a perturbation framework [12] and has been implemented nonperturbatively in nuclear physics [4]. A difficulty may arise in atomic problems with more than two electrons for which nonlocal exchange terms arise. Local approximations to the exchange potential are available, but their reliability in collision problems is uncertain [13].

E. Multiphoton Ionization

If the field due to moving nuclei in Eq. (2) or (9) is replaced by an external electric field, e.g., due to a laser,

$$v = \mathbf{F}(t) \cdot \mathbf{r} \quad (11)$$

we can use the same methods to study multiphoton ionization [14]. The currently fashionable problems of collisions in electric and magnetic field can all be readily formulated and should be tractable if appropriate numerical methods are available [15].

F. Remarks on Stationary Problems

In time-dependent theory, stationary scattering problems, e.g., Eq. (5), are solved by scattering wavepackets [3]. It might be asked what one loses by not using the well-known separation of time in Eq. (1) when H is independent of t ,

$$\Psi(q, t) = \Phi(q) \exp(-iEt) \quad (12)$$

The answer is that one should not lose anything. The stationary eigenstates may be recovered from any time-dependent solution by a Fourier transformation. Most stationary methods for scattering problems involve the diagonalization of large matrices while it may be shown that the iterative method of diagonalization is equivalent to propagating Eq. (1) for imaginary times [16]. What one gains is to avoid the sometimes knotty problem of asymptotic boundary conditions.

2. Construction of Propagators

A. General Remarks: Stability

It is convenient to separate the discretization of Eq. (1) in space from the explicit construction of an operator which propagates the solution in time. Let time be divided into intervals of duration τ so that

$$\Psi(t + \tau) = G(t, \tau) \Psi(t) \quad (13)$$

where G is given by Dyson's exponential [17]

$$G(t, \tau) = T \exp \left[-i \int_t^{t+\tau} H(s) ds \right] \quad (14)$$

So far, our equations are exact. Before approximating Eq. (14) we must introduce the question of stability.

The solution of parabolic equations is notoriously plagued by instabilities due to the amplification of short-wavelength noise (in the space domain). Suppose we calculate the effect of an approximate G , say G_1 , on a test function [18]

$$\chi(\mathbf{k}) = \exp(i\mathbf{k} \cdot \mathbf{r}) \quad (15)$$

If

$$G_1\chi(\mathbf{k}) \cong \mathcal{A}(\mathbf{k})\chi(\mathbf{k}) \quad (16)$$

we call $\mathcal{A}$ the “amplification factor.” If

$$|\mathcal{A}(\mathbf{k})| \leq 1 \text{ for } k < 2\pi/\Delta x \quad (17)$$

the algorithm G_1 is “unconditionally stable” in the sense that noise is damped out or at least not amplified. The minimum wavelength Δx is a typical mesh spacing. The discussion of stability is simplified if one is familiar with the language of quantum mechanics. If G_1 is unitary [17],

$$G_1^\dagger G_1 = 1 \quad (18)$$

it follows from Eq. (16) that

$$|\mathcal{A}(\mathbf{k})| = 1 \quad (19)$$

and stability follows. Thus, we shall try to construct propagators as nearly unitary as possible. This result harmonizes with most experience in solving quantal problems, which suggest that it is desirable to conserve probability, i.e. to use unitary propagators. We now turn to approximation schemes.

B. Taylor Series

To expand the ordered exponential termwise makes little sense as a computational algorithm, since one would have to further subdivide the interval τ . It is always preferable to use a simpler method with a smaller interval. However, a family of unitary approximations is suggested by the Magnus expansion [19]

$$G(t, \tau) = \exp(-iA_1 - iA_2 + \dots) \quad (20)$$

where

$$A_1 = \int_t^{t+\tau} H(s)ds, \quad A_2 = \frac{1}{2i} \int_t^{t+\tau} ds_1 \int_t^{s_1} ds_2 [H(s_1), H(s_2)] \quad (21)$$

If H varies slowly with time we can neglect A_2 and set

$$A_1 \cong \tau H(\hat{t}) \quad (22)$$

where $\hat{t} = t + 1/2\tau$. For matrices[†] of large dimension, Eq. (20) must be evaluated

[†] Operators are always represented by matrices.

by the power series expansion, the convergence being at worst comparable with $\exp(\tau k_{\max}^2)$. This method, retaining five or six terms, has been used successfully in nuclear problems [20].

C. Predictor–Corrector Methods

These techniques may be regarded as more efficient rearrangement of a Taylor series. One has to carry out a series of operations of the form

$$\psi_j = a_j \psi_{j-1} + b_j H \psi_{j-1} \quad (23)$$

They have a very large literature [21]. The drawback of these “explicit” methods in atomic problems is that one must go to a high order in τ to ensure stability. Thus, we turn to “implicit” methods in the next subsection.

D. The Peaceford–Rachman Algorithm

In an implicit method, one has to invert a matrix to obtain the wavefunction at the next step. Thus, Eq. (14) may be approximated by

$$G_1 = (1 + \frac{1}{2}iA_1)^{-1}(1 - \frac{1}{2}iA_1) \quad (24)$$

which is exactly unitary. If Eq. (22) is used, the error in A_1 is $O(\tau^3)$. If $H(s)$ is constant over $t < s < t + \tau$, $A_2/\tau^2 = 0$: Thus A_2 is in fact $O(\tau^3)$. Finally, the error in Eq. (24) is also $O(\tau^3)$.

The construction of the inverse in Eq. (24) can be greatly simplified by decomposing the Hamiltonian, e.g. in two dimensions x, y , we write

$$H = H_x + H_y, H_x = T_x + \eta V, H_y = T_y + (1 - \eta)V \quad (25)$$

where T, V are kinetic, potential energy operators. Then, H_x, H_y are local operators in y, x . If we expand the following operator to $O(\tau^2)$,

$$G_{PR} = L_y^{-1} L_x^{-1} L_x^* L_y^*, L_w + 1 + \frac{1}{2}i\tau H_w(\hat{t}) \quad (26)$$

we find

$$G_{PR} = 1 - iA_1 - \frac{1}{2}A_1^2 + \dots \quad (27)$$

so that G_{PR} is correct to $O(\tau^2)$. If the operators in Eq. (26) were nested as $yyxx$ or $xxyy$, G_{PR} would be exactly unitary, but the error would become $O(\tau^2)$. If H_x and H_y do not commute, G_{PR} is not unitary but $|\mathcal{A}| \equiv 1$ so that stability is assured. In practice, unitarity is easily satisfied to high accuracy. The Peaceford–Rachman algorithm, Eq. (26), has a large literature [18, 22].

If we formally extend Eq. (26) into three dimensions

$$G_{PR} = L_z^{-1} L_y^{-1} L_x^{-1} L_x^* L_y^* L_z^* \quad (28)$$

the expansion (27) continues to hold, so that the error is $O(\tau^3)$. Further $|\mathcal{A}| \equiv 1$, so that stability carries over. It should be noted that for the diffusion equation (in real time) in 3D, stability breaks down [18].

The expression (24) may be regarded as the first in a family of Padé approximants which represent G to any order in τ [23]. However, one should be aware that any finite approximant has nonphysical singularities in complex coordinate space. This feature has consequences in numerical analysis: consider the inhomogenous equation

$$L_x \psi = f. \quad (29)$$

If $V \rightarrow 0$ as $x \rightarrow \infty$, the solutions of the homogenous equation have the asymptotic form

$$\psi \sim ae^{-\lambda x} + be^{\lambda x}, \lambda^2 = 2/\tau. \quad (30)$$

Numerical solutions must discard the exponentially growing term.

3. Discretization in Space

A. Finite Differences

We begin with the orthodox finite difference method applied to Eq. (29), with

$$H_x = T_x + V(x), T_x = -\frac{1}{2} \frac{\partial^2}{\partial x^2} \quad (31)$$

Introducing a uniform grid [21]

$$x_n = nh, 0 \leq n \leq N \quad (32)$$

Eq. (29) is replaced by the matrix relation

$$\mathbf{L}\psi = \mathbf{f}, \mathbf{L} = \mathbf{I} + \frac{1}{2}i\tau(\mathbf{T} + \mathbf{V}) \quad (33)$$

where $\psi_n = \psi(x_n)$, etc., and

$$(\mathbf{V}\psi)_m = V(x_m)\psi_m, (\mathbf{T}\psi)_m = -\frac{1}{2h^2}(\psi_{m+1} - 2\psi_m + \psi_{m-1}). \quad (34)$$

Boundary conditions must also be considered. If

$$xV \rightarrow -Z(<0) \text{ as } x \rightarrow 0 \quad (35)$$

we must have

$$\psi_0 = 0. \quad (36)$$

Usually $\psi \rightarrow 0$ as $x \rightarrow \infty$, so that one condition of the family

$$\left(\frac{\partial}{\partial x}\right)^K \psi = 0, x = Nh \quad (37)$$

is required; $K = 2$ appears to give particularly stable results [24].

In Sec. 3.D we shall discuss how instabilities arise from the Coulomb singularity. Instabilities of another type can appear if the grid spacing is nonuniform, but nonuniform grids are hardly mentioned in the literature.

B. Finite Elements in One Dimension

In finite element methods, physical problems are solved by expanding in “well-behaved” functions confined to localized regions of space [25]. We shall confine our attention to square-integrable, continuous functions (though first derivatives may be discontinuous). The simplest approach (another is given in Sec. 3.C) is to consider an expansion in real, square-integrable functions,

$$\psi(x) = \sum \psi_n \alpha_n(x) \quad (38)$$

where

$$\int \alpha_m \alpha_n dx = S_{mn}, \quad \int \alpha_m L_x \alpha_n dx = L_{mn} \quad (39)$$

The operator L_x was introduced in Eq. (26). We note that Eq. (29) is equivalent to minimizing

$$M[\psi, f] = \int (f - L_x \psi)^2 dx \quad (40)$$

with respect to f . Alternatively,

$$\frac{\delta M}{\delta f} = 2 \int \delta f (f - L_x \psi) dx = 0 \quad (41)$$

If we make the choice $\delta f = \epsilon \alpha_m$, Eq. (41) becomes

$$\int \alpha_m (f - L_x \psi) dx = 0 \quad (42)$$

These are the equations of the “collocation” or Galerkin method [25]. Expanding f and ψ according to Eq. (38) we obtain in place of Eq. (29),

$$\mathbf{L}\psi = \mathbf{S}f \quad (43)$$

Apart from the overlap matrix, this is similar to Eq. (33), but a global error bound is provided by Eq. (40). Thus, we generally expect collocation methods to be more stable than finite difference methods [26].

The simplest localized, continuous functions are the piecewise linear “tents,”

$$\begin{aligned} \alpha_m(x) &= (x - x_{m-1})/(x_m - x_{m-1}) x_{m-1} < x < x_m \\ &= (x_{m+1} - x)/(x_{m+1} - x_m) x_m < x < x_{m+1} \\ &= 0 \quad x > x_{m+1}, x < x_{m-1} \end{aligned} \quad (44)$$

The overlap matrix elements are given by

$$S_{nn} = 2/3(x_{n+1} - x_{n-1}), S_{nn+1} = 1/6(x_{n+1} - x_n) \quad (45)$$

and the kinetic energy matrix elements in Eq. (31) by

$$T_{nn} = \frac{1}{2} \left(\frac{1}{x_n - x_{n-1}} + \frac{1}{x_{n+1} - x_n} \right), T_{nn+1} = -\frac{1}{2} \left(\frac{1}{x_{n+1} - x_n} \right) \quad (46)$$

The potential matrix elements can be done analytically, in simple cases, or numerically; a two-point Gaussian quadrature in each interval $x_{m-1} < x < x_m$ is usually adequate. Formulae for more complicated coordinate systems are easily worked out. It is clear that all matrices in this method are tridiagonal. This is the reason for the separation of Eq. (25) in the first place. No problems arise if the nodes are nonuniformly spaced: in atomic physics one usually likes to have a fine mesh spacings near the nuclei and broader spacings far away.

With the functions of Eq. (44) Eq. (38) leads to

$$\psi(x_m) = \psi_m \quad (47)$$

so that a close relation to the finite difference method is apparent. The next step would be to include parabolae

$$\omega_n = (x - x_{m-1})(x_m - x) \quad (48)$$

which would generate banded matrices five elements wide.

In contrast to the finite-difference method the variational principle (41) also takes care of the boundary conditions. Equations (43) are well-posed as they stand. For potentials of the form (35), the first nodal point $x_0 = 0$ and the first tent is centered on x_1 .

Because of the exponentially growing terms in Eq. (30) the matrix L in Eq. (33) is not inverted directly [18]. Since Eq. (33) is a three-term recurrence relation, two independent solutions are propagated from $n = 0$ and $n = N$ and matched at some intermediate points. If x_N is chosen too large the matching may be ill-conditioned.

C. Finite Elements in Many Dimensions

Factorization into 1D problems (25), (26) has its drawbacks, since one is limited to rectangular grids. Suppose one wants to place more finite elements around a nucleus in 2D or 3D: it is obviously more efficient to cluster the elements around a point than along an axis. The formal equation (43) continues to hold, but the matrices have a more complicated structure.

The tents (44) can be generalized to piecewise linear functions in 2D by dividing the plane into overlapping hexagons. Inside each hexagon is one node which is a vertex of six other hexagons. Each element is unity at the interior node and zero on the edges. Banded matrices seven elements wide result. Vastly more elaborate schemes have been implemented in 2D and 3D [25].

Finite elements, particularly in many dimensions, may be approached through physical intuition rather than mathematics. One asks whether the physical system can be broken down into a series of simpler systems connected only at a finite number of points. Thus, a continuous elastic plate can be approximated by many smaller plates bolted together. Suppose Eq. (1) describes the motion of an electron in a time-dependent field. A possible model is that the electron is scattered by a series of discrete centers. To formalize this model we divide space

into a series of boxes B_I in each of which we apply the variational principle that

$$F[\psi] = \int \psi^* H \psi d\tau - \frac{1}{2} i \int \left(\psi^* \frac{\partial \psi}{\partial t} - \psi \frac{\partial \psi^*}{\partial t} \right) d\tau + \frac{1}{4} \int \left(\psi^* \frac{\partial \psi}{\partial n} - \psi \frac{\partial \psi^*}{\partial n} \right) d\sigma. \quad (49)$$

The Hamiltonian is taken to have the form Eq. (2); $d\tau$, $d\sigma$ are volume and surface elements of B_I , and $\partial/\partial n$ is the normal derivative.

To obtain a finite set of equations, we expand

$$\psi I(r) = \sum_{n=1}^M f_n^I \alpha_n(r) \quad (50)$$

whence Eq. (49) yields

$$i \frac{d}{dt} \mathbf{f}_I = (\mathbf{H}_I + \mathbf{K}_I) \mathbf{f}_I. \quad (51)$$

The skew-symmetric matrix $\mathbf{K}_I$ arises from the boundary term and allows for flux in or out of B_I . It is more convenient to work with values of the wavefunction at nodal points on the boundary, say

$$\psi_I(\mathbf{r}_m) = g_m^I = \sum f_n^I \alpha_n(\mathbf{r}_m). \quad (52)$$

Letting $\alpha_n(\mathbf{r}_m) = A_{mn}^I$, Eq. (51) becomes

$$i \frac{d}{dt} \mathbf{g}_I = \{\mathbf{A}_I (\mathbf{H}_I + \mathbf{K}_I) \mathbf{A}_I^{-1}\} \mathbf{g}_I. \quad (53)$$

The set $\mathbf{g}_m^I$ could also include normal derivatives of ψ_I . If contingent boxes I , J share the same nodes, we can add equations of continuity

$$\mathbf{g}_I = \mathbf{C}_{IJ} \mathbf{g}_J. \quad (54)$$

Together, Eqs. (53) and (54) are overdetermined, but this is easily remedied, e.g., by taking linear combinations of equivalent expressions.

An interesting feature of Eq. (53) is that it takes proper account of flux out of the overall volume of the calculation, B . Thus B can be chosen smaller than in other methods. Since there is no dynamical mechanism for returning particles to B once they leave, the predicted probability of ionization (flux to infinity) must be an upper bound,

$$P_{\text{ion}}(B) \geq P_{\text{ion}}(\infty). \quad (55)$$

This result has an affinity to the Wigner-Keck variational principle for reaction rates [27].

D. Stability in the Presence of Coulomb Singularities

We shall now discuss in detail how numerical instabilities arise near a Coulomb singularity and how the finite element method removes them. Consider Eq. (29) again, where L_x can be written

$$L_x = a_0 + a_1 C, \quad C = \left(\frac{d^2}{dx^2} + \frac{2Z}{x} \right). \quad (56)$$

The inverse operator

$$L_x^{-1} = a'_0 + a'_1 C + a'_2 C^2 + \dots \quad (57)$$

Now each function ψ or f should satisfy the “cusp condition” near $x = 0$, but in reality

$$f = bx(1 - Zx) + \Delta f + O(x^2) \quad (58)$$

where Δf is the rounding error. Then on applying Eq. (57) K times,

$$\psi \Delta \sim \Delta f (\tau/x)^K \quad (59)$$

which grows rapidly if $x \sim 0$. If $\Delta f = 0$, $Cf = O(x)$ only. The divergence is much worse if the nucleus is moving between time steps.

To see what happens in the finite element method, we consider the simplest case of uniformly spaced nodes. If the functions (44) are renormalized so that the diagonal overlaps are unity, it is readily shown that Eq. (43) has a form close to the finite difference equations (33) and (34) except that f and V are replaced by “smoothed” functions

$$\hat{f}_n = (SF)_n = \frac{1}{4}(f_{n+1} + 2f_n + f_{n-1}), \quad \hat{V}_n = (SV)_n. \quad (60)$$

We now consider a hierarchy of basis sets

$$\alpha_n^{(K+1)} = (S^{K/2} \alpha)_n \quad (61)$$

such that for the K th set, S in Eq. (60) is replaced by S^K . As K increases, the potential becomes smoother and instabilities less severe. Thus, the finite difference method can be regarded as the zeroth member, and the simple tent-function finite element method as the first member of a sequence of increasingly stable algorithms. Expressed otherwise, the more overlap matrix elements are nonzero, the better the method.[†]

4. Conclusions

We have now applied finite element methods successfully to 2D models of $e + H$ and $p + H$ scattering [28]. Extensions to other problems surveyed in Sec. 1 are underway.

[†] Within reason: different problems arise if the rows of S become linearly dependent.

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